Ex. No.:4	PERSONALIZED RECOMMENDATION SYSTEM	URK23CS1167
17.09.2026		

Aim:

To develop a personalized recommendation system using GPT-4 that analyzes user preferences and generates, ranks and explains suitable recommendations based on the user's profile, interests, skill level and goals.

Problem Statement:

Design and develop an intelligent personalized recommendation system using GPT-4. The application should collect user information and preferences, analyze the user's requirements, generate personalized recommendations, rank them according to suitability, provide explanations for the recommendations and refine the results based on user feedback.

Software Requirements:

Software	Version
Python	3.13.5
Ollama	0.40.13
VS Code	1.144.0
Pandas	3.1.0
Sentence Transformers	6.2.0
Flask	3.1.3
LLM Model	Llama 3.2 (3B) via Ollama

Input Parameters:

Parameters	Description
Name	User's name. Example: Monick
Age	User's age. Example: 21
Background	Education/work background. Example: Final-year B.Tech Computer Science student
Interests	Comma-separated interests. Example: Artificial Intelligence, Python, Cloud
Skill Level	Beginner, Intermediate or Advanced
Preferred Category	Any, Course, Project, Book or Certification
Goal	What the user wants to achieve. Example: Build real AI applications and get placed

Parameters	Description
Preferences	Free text. Example: I prefer practical, hands-on learning with projects
Number of items	How many recommendations to return (1 – 10). Example: 5

System Architecture:

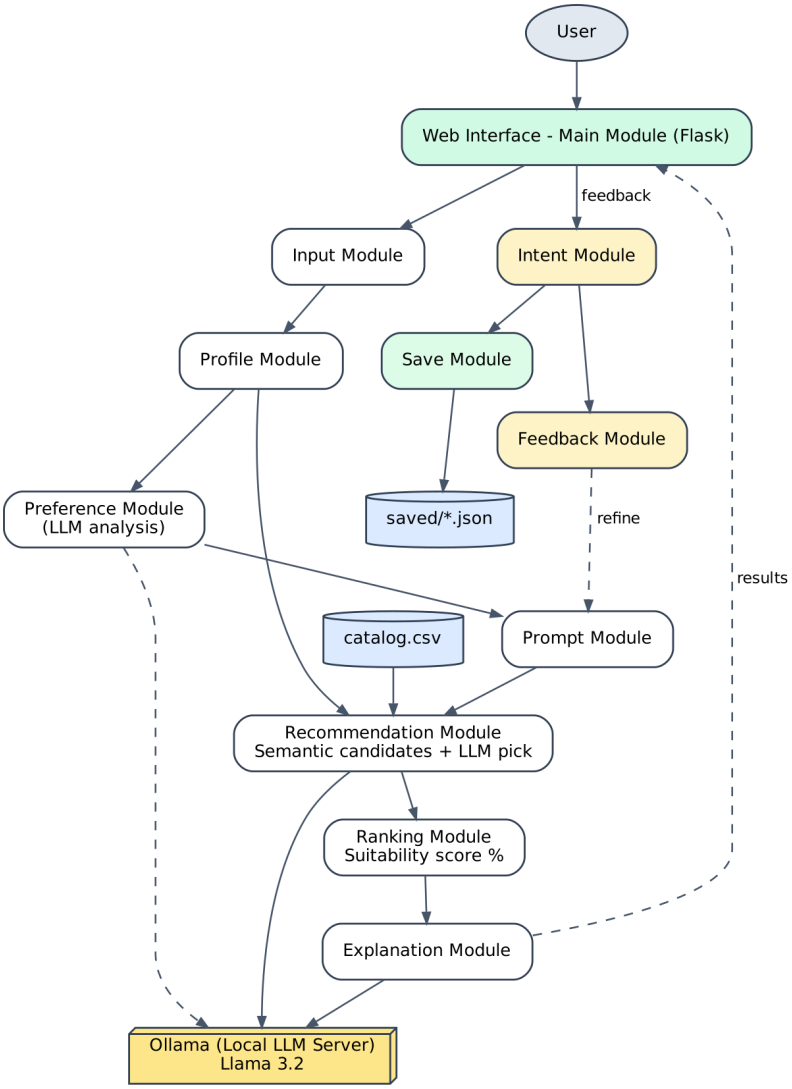


Fig. 4.1 System architecture

Module Design:

Module	Responsibility
Input Module	Collects and validates the user details from the web form.
Profile Module	Organizes the inputs into a structured user profile with a one-line summary.
Preference Module	Uses the LLM to analyze free-text preferences: key interests, learning style, focus areas and things to avoid.

Module	Responsibility
Prompt Module	Builds a structured prompt with role, profile, preference analysis, candidate items, task and JSON output format.
Recommendation Module	Retrieves candidate items from catalog.csv using semantic search and lets the LLM pick the best items for the user.
Ranking Module	Calculates a suitability score: 50% similarity + 20% level match + 15% learning-style match + 15% LLM rank, with a penalty for avoided topics.
Explanation Module	Generates a simple explanation of why each item suits the user.
Feedback Module	Updates interests, learning style and topics to avoid from user feedback and produces refined recommendations.
Intent Module	Classifies the user's message as REFINE, EXPLAIN, SAVE or NEW.
Save Module	Saves the profile and ranked recommendations to a JSON file.
Main Module	Flask application (app.py) that connects all modules into one pipeline.

Implementation:

• GitHub Link - https://github.com/MONICK10/GPT4-Exp4-Personalized-Recommendation-System
• LinkedIn Post Link - <add LinkedIn post link in config.json>
• Run: `python app.py` → `http://localhost:5004`

Output Screenshots:

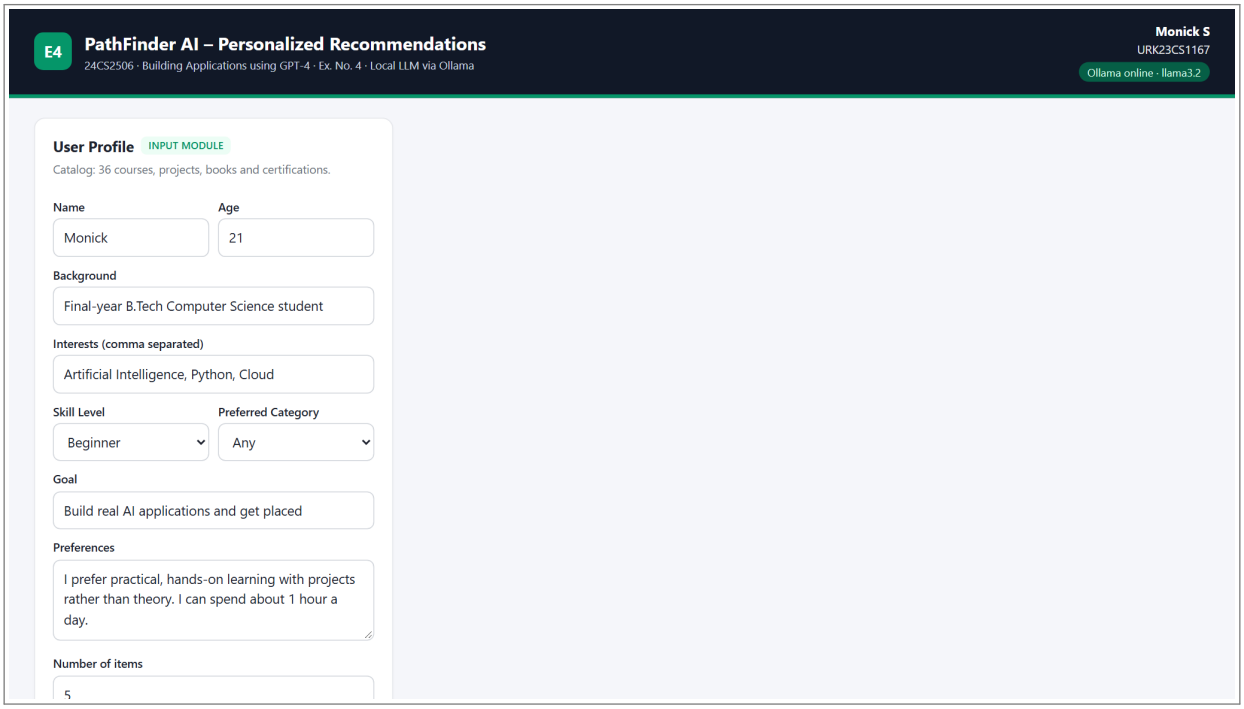


Fig. 4.2 Home Screen

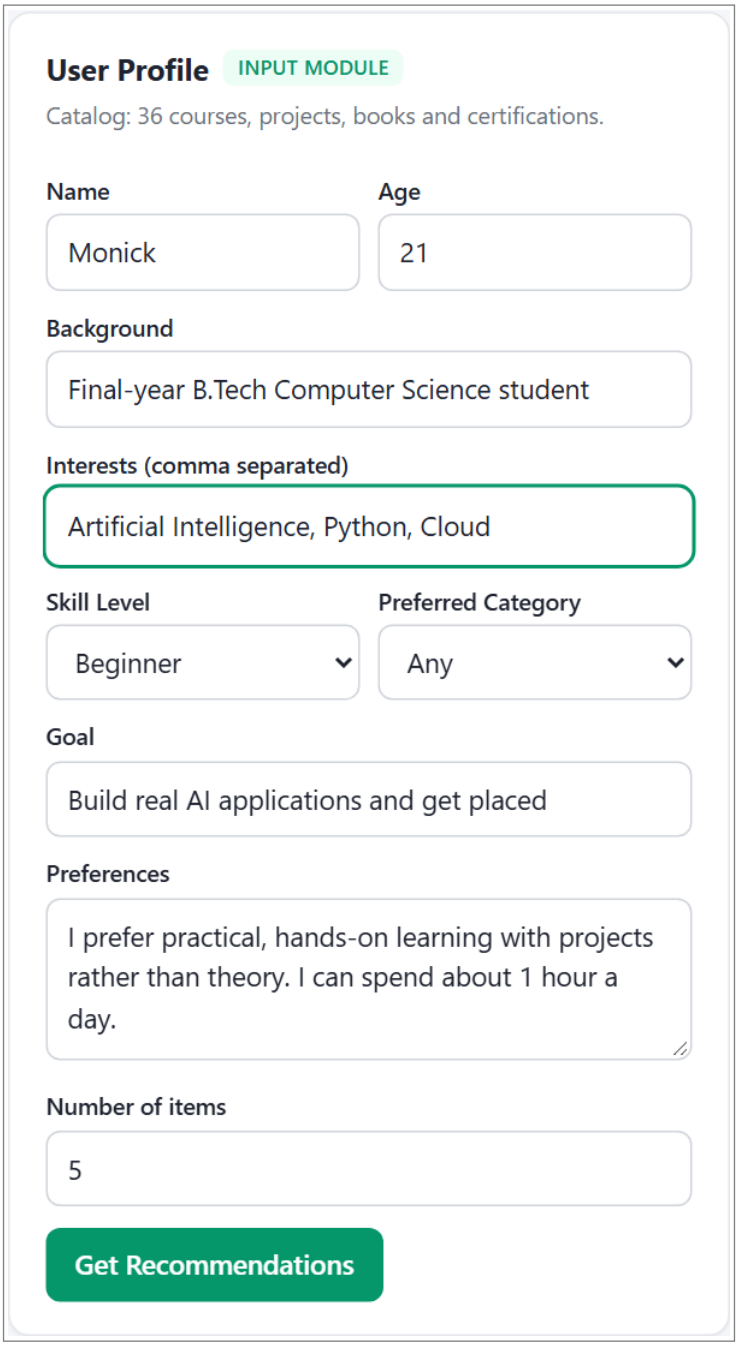


Fig. 4.3 User Profile Input

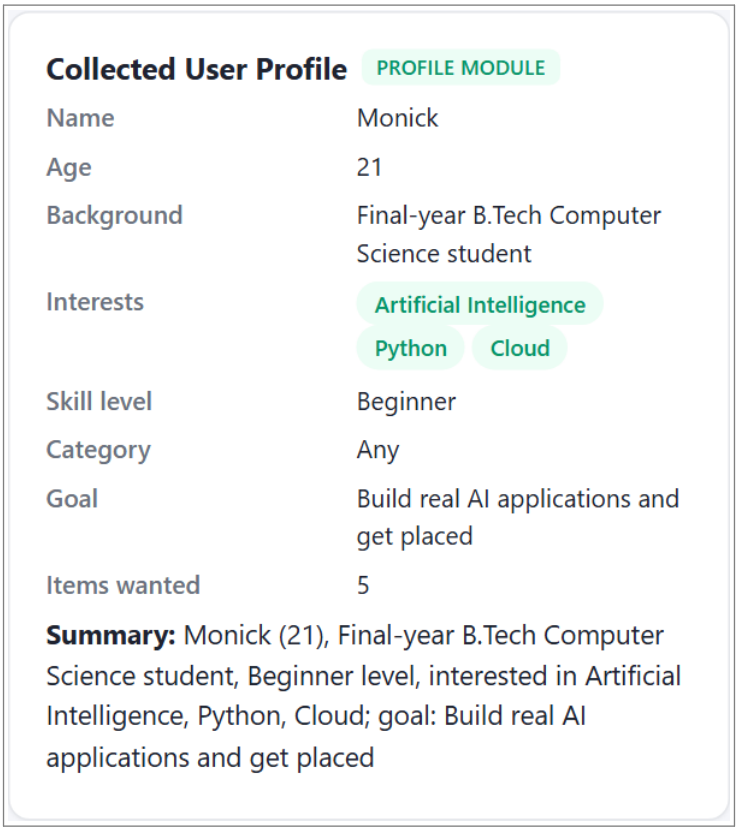


Fig. 4.4 Collected User Profile

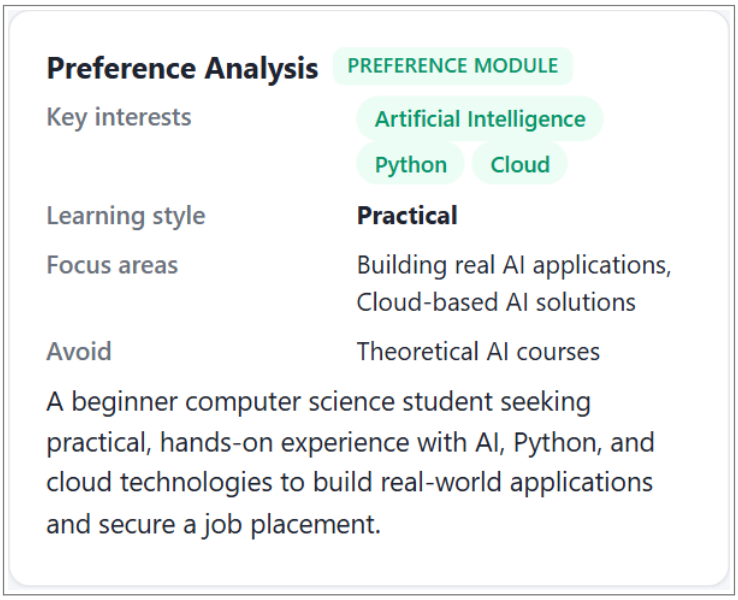


Fig. 4.5 Preference Analysis

Prompt Generation PROMPT MODULE

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ROLE: You are a personalised learning recommendation assistant.

USER PROFILE:
- Name: Monick
- Age: 21
- Background: Final-year B.Tech Computer Science student
- Interests: Artificial Intelligence, Python, Cloud
- Skill level: Beginner
- Preferred category: Any
- Goal: Build real AI applications and get placed
- Preferences: I prefer practical, hands-on learning with projects rather than theory. I can spend about 1 hour a day.

PREFERENCE ANALYSIS:
- Key interests: Artificial Intelligence, Python, Cloud
- Learning style: Practical
- Avoid: Theoretical AI courses

CANDIDATE ITEMS (id | title | category | level | type | description):
C05 | Generative AI with Python | Course | Intermediate | Practical | Build apps with LLMs, prompt engineering,
embeddings, RAG and local models using Ollama.
B02 | Automate the Boring Stuff with Python (Book) | Book | Beginner | Practical | Practical Python programming for
automating everyday tasks.
B01 | Hands-On Machine Learning (Book) | Book | Intermediate | Mixed | Practical ML and deep learning with Scikit-
Learn, Keras and TensorFlow.
C12 | Backend APIs with Flask and FastAPI | Course | Intermediate | Practical | Build REST APIs in Python,
authentication, databases and deployment.
B03 | Python Crash Course (Book) | Book | Beginner | Practical | Fast-paced introduction to Python with three hands-on
projects.
C01 | Python Programming for Beginners | Course | Beginner | Practical | Learn Python basics: variables, loops,
functions, files and small hands-on projects.
P02 | AI Chatbot using RAG Project | Project | Intermediate | Practical | Create a document Q&A chatbot with
embeddings, vector search and an LLM.
```

Fig. 4.6 Prompt Generation

LLM Recommendations RECOMMENDATION MODULE

Items chosen by the LLM from the semantic-search candidates (Sentence Transformers (all-MiniLM-L6-v2)).

1. **Automate the Boring Stuff with Python (Book)** (B02)
   Practical Python programming for automating everyday tasks aligns with Monick's interest in practical learning.
2. **Python Programming for Beginners** (C01)
   Learn Python basics with hands-on projects is a good fit for Monick's skill level and learning style.
3. **AI Chatbot using RAG Project** (P02)
   Create a document Q&A chatbot with embeddings, vector search and an LLM aligns with Monick's interest in AI and practical learning.
4. **Python Crash Course (Book)** (B03)
   Fast-paced introduction to Python with three hands-on projects meets Monick's preference for practical learning.
5. **Backend APIs with Flask and FastAPI** (C12)
   Build REST APIs in Python, authentication, databases and deployment aligns with Monick's interest in Cloud and practical learning.

Fig. 4.7 GPT-4 Recommendation

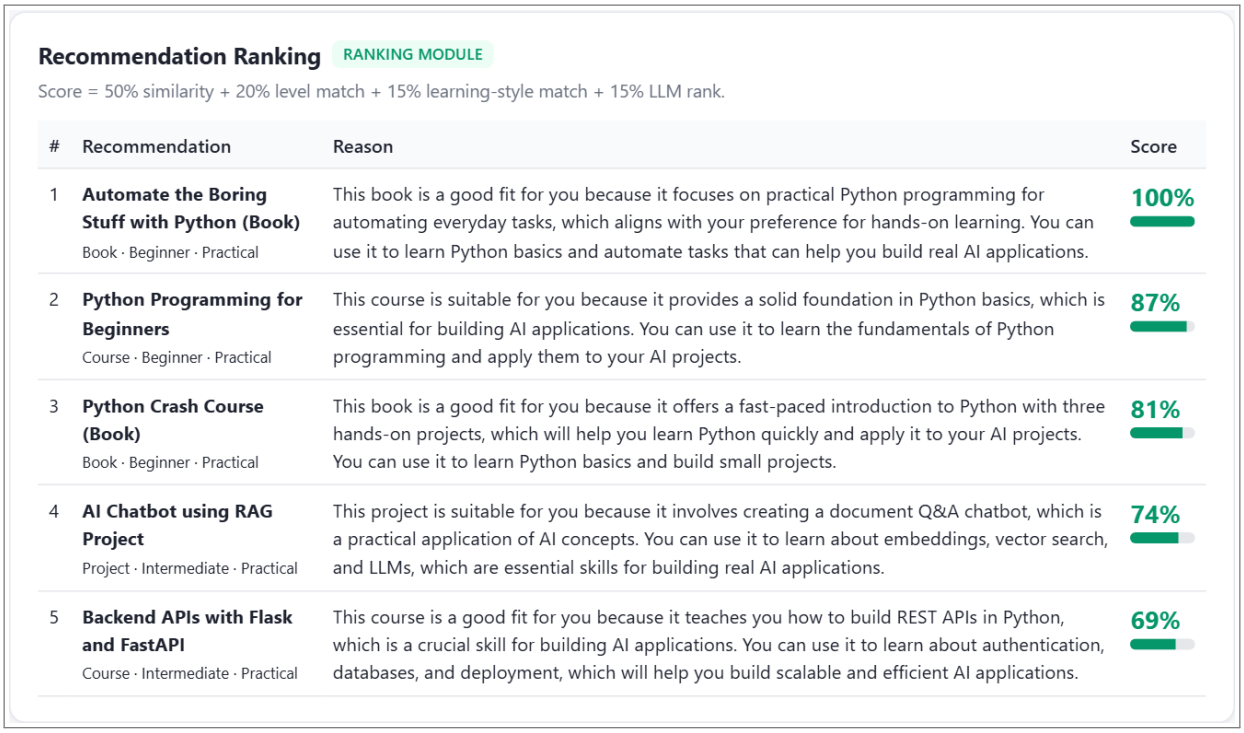


Fig. 4.8 Recommendation Ranking

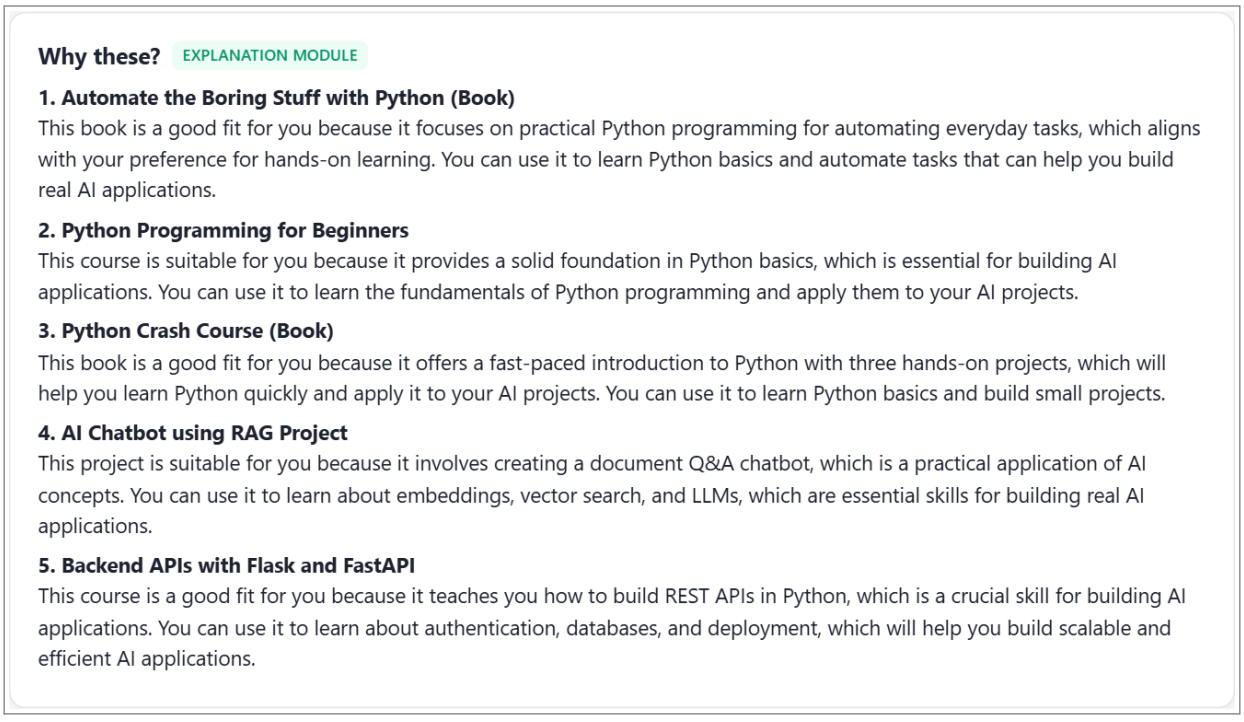


Fig. 4.9 Recommendation Explanation

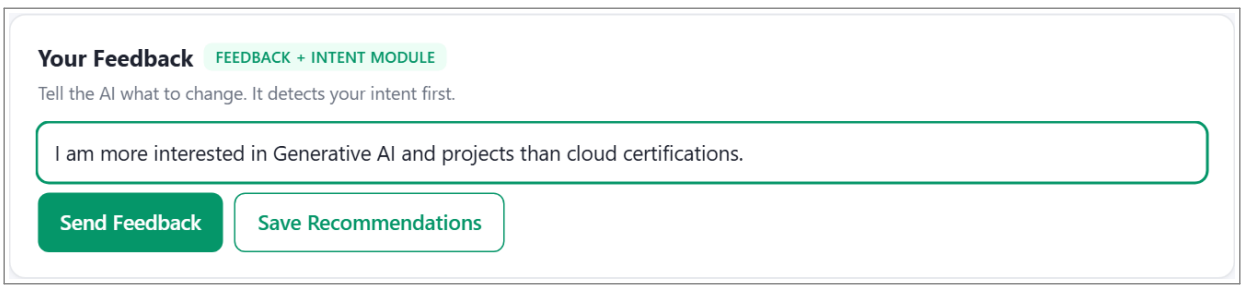


Fig. 4.10 User Feedback

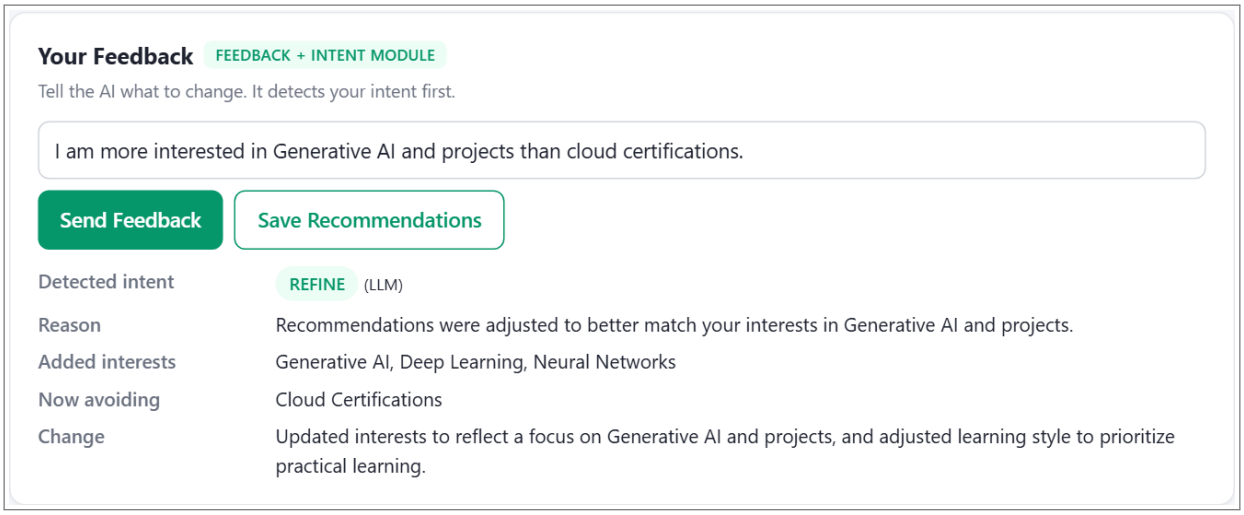


Fig. 4.11 Intent Detection

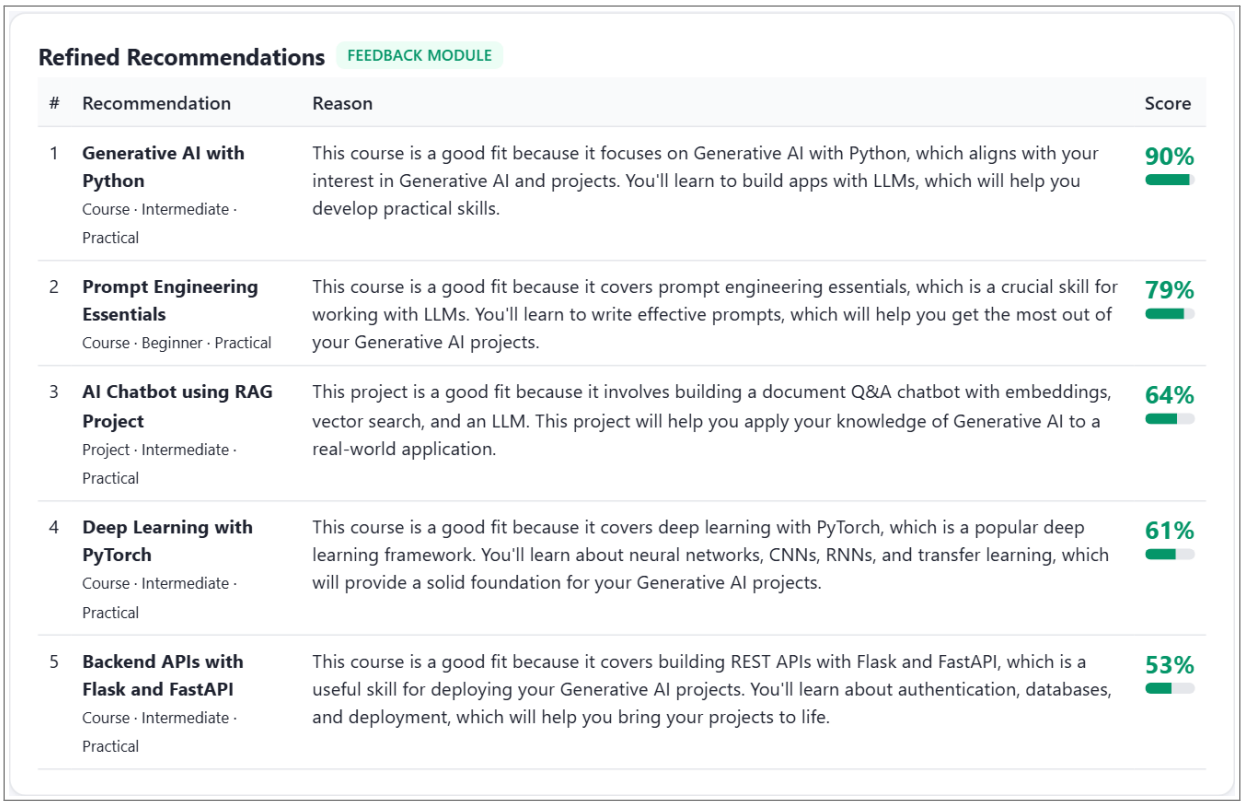


Fig. 4.12 Refined Recommendations

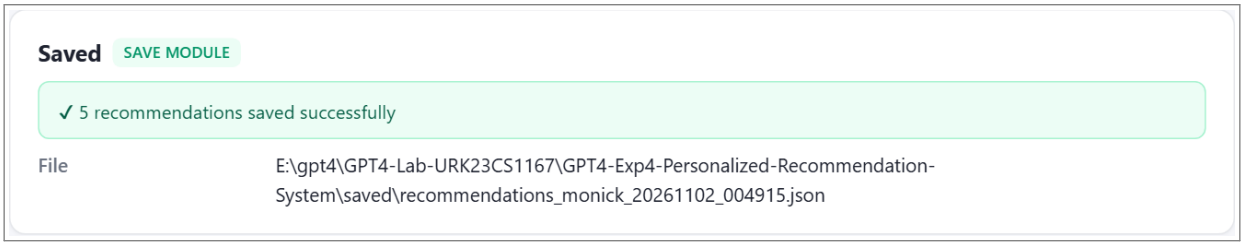


Fig. 4.13 Recommendation Saved Successfully

Result:

Thus, a personalized recommendation system using GPT-4 was successfully developed. The system collects user information, analyzes preferences, generates personalized recommendations, ranks and explains the recommendations, accepts user feedback and produces refined recommendations.